1. Project Title

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Reg No – 24BKT0106

Web Page Link: file:///Users/sidharthjain/Assignment2.html

GitHub Repository: https://github.com/SidharthJain2006/24BKT0106

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1. Project Title

Smart Calculator

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2. Objective

The Smart Calculator is an online calculator meant for doing basic math calculations, including additions, subtractions, multiplications, and divisions. The purpose of this project is to enable users to do their calculations fast and understand how JavaScript, HTML, and CSS work.

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3. Features

- Ability to add, subtract, multiply and divide.
- Button marked C for clearing everything entered so far.
- Delete button removes the last item entered.
- A display screen for showing inputted data, operator and last result.
- User-friendly interface.
- Black background for the calculator with contrasting buttons.

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4. Technologies Used

- HTML – for the calculator structure.
- CSS – for styling and layout of calculator buttons and display.
- JavaScript – to implement all functionalities, event handling, and calculations.
- Git and GitHub for version control and submission.
- Visual Studio Code(Code Editor)

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5. Documented Steps

1. Project Setup
    - Created the Assignment2.html file with a basic calculator layout using buttons and display divs.
2. CSS Styling
    - Styled the calculator with a dark theme using background-colour, border-radius, and grid layout for the buttons.
3. JavaScript Logic
    - Implemented functions to handle:
        - Number input (appendNumber)
        - Operator selection (chooseOperator)
        - Computation (compute)
        - Delete last character (deleteLast)
        - Clear all (clear)
4. Button Functionality
    - Added click event listeners to all buttons using addEventListener.
    - Ensured operators, numbers, clear, delete, and equals work correctly.
5. Testing
    - Tested addition, subtraction, multiplication, and division.
    - Verified delete and clear buttons.

    - Checked display updates and hover effects.
6. Final Touch
    - Adjusted CSS for a sleek dark calculator look.
    - Ensured buttons are visible and the calculator is easy to use.

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6. Learning Reflections (10 Points)

1. Learned how to create an HTML structure that would support interaction for web apps.
2. Applied CSS style, including grids and dark mode themes.
3. Used DOM manipulation methods such as getElementById and querySelectorAll. Learned to handle user input with addEventListener.
4. Learnt how to capture user inputs using addEventListener method.
5. Applied arithmetic calculations in JavaScript.
6. Converted strings to numbers using parseFloat method in JavaScript.
7. Practiced updating the contents of HTML using JavaScript..
8. Applied functionality and design to create a friendly calculator

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7. Importance of the Project

Significance of the Smart Calculator Project:

This project is crucial in demonstrating the connection between the fundamental ideas of programming and their real-world application to user interface design. It improves one's proficiency in programming languages such as JavaScript, HTML, and CSS. In addition, it enables comprehension of the process through which input provided by the user is dynamically generated.

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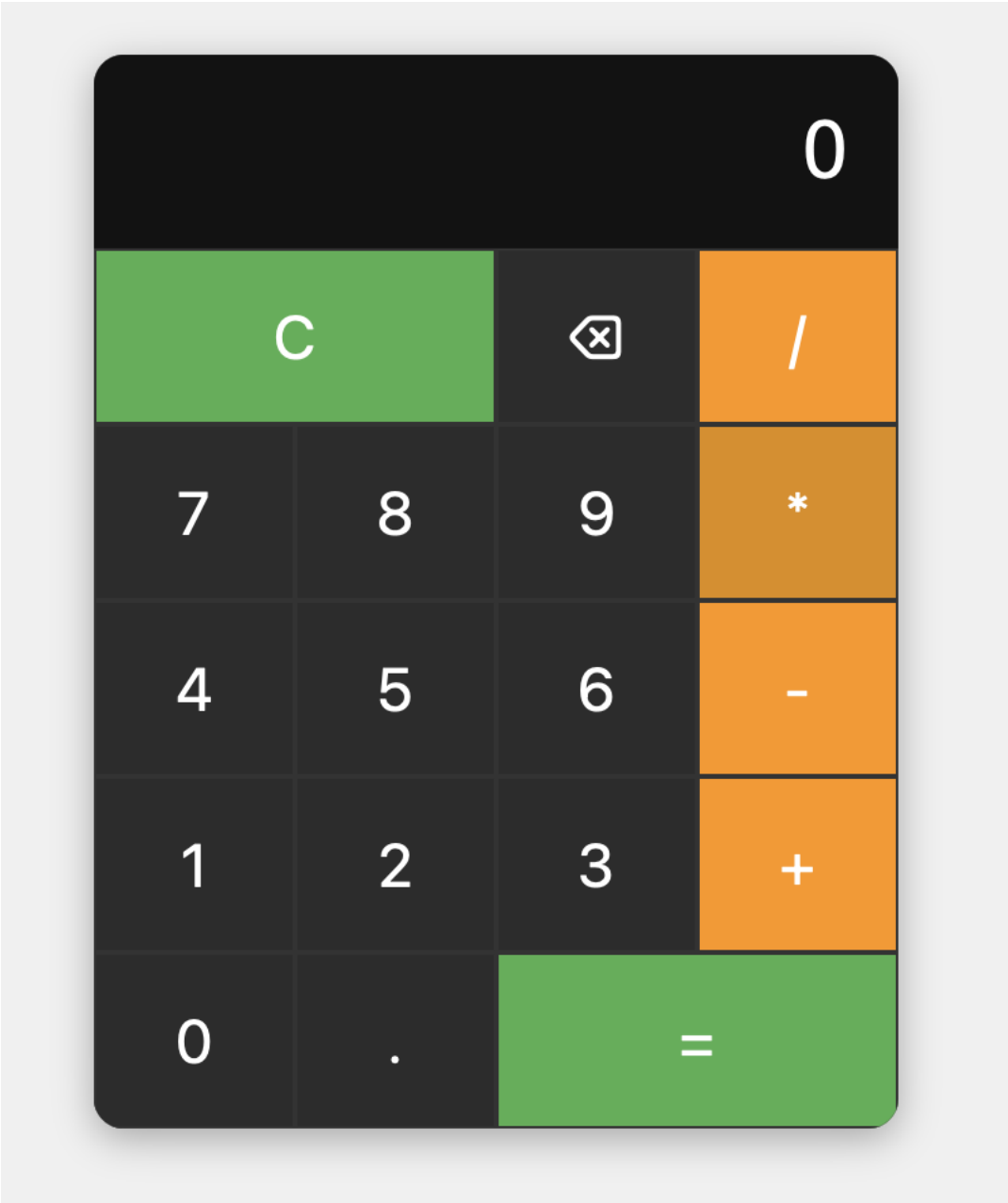
8. Screenshots

1. Initial Calculator Display: Shows 0 on the screen with all buttons visible.
2. After Entering Numbers: Demonstrates typing numbers into the calculator.
3. After Performing an Operation: Shows the result after clicking =.
4. After Using Clear/Delete: Demonstrates C or functionality.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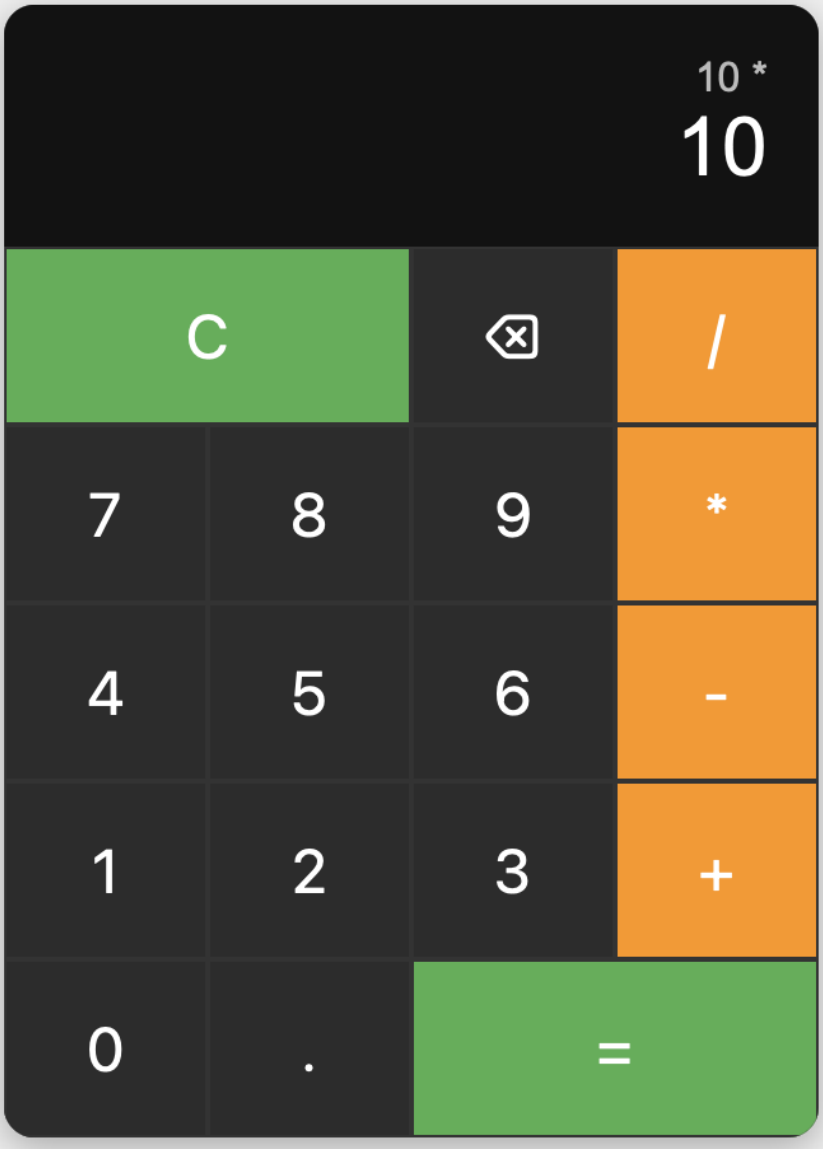
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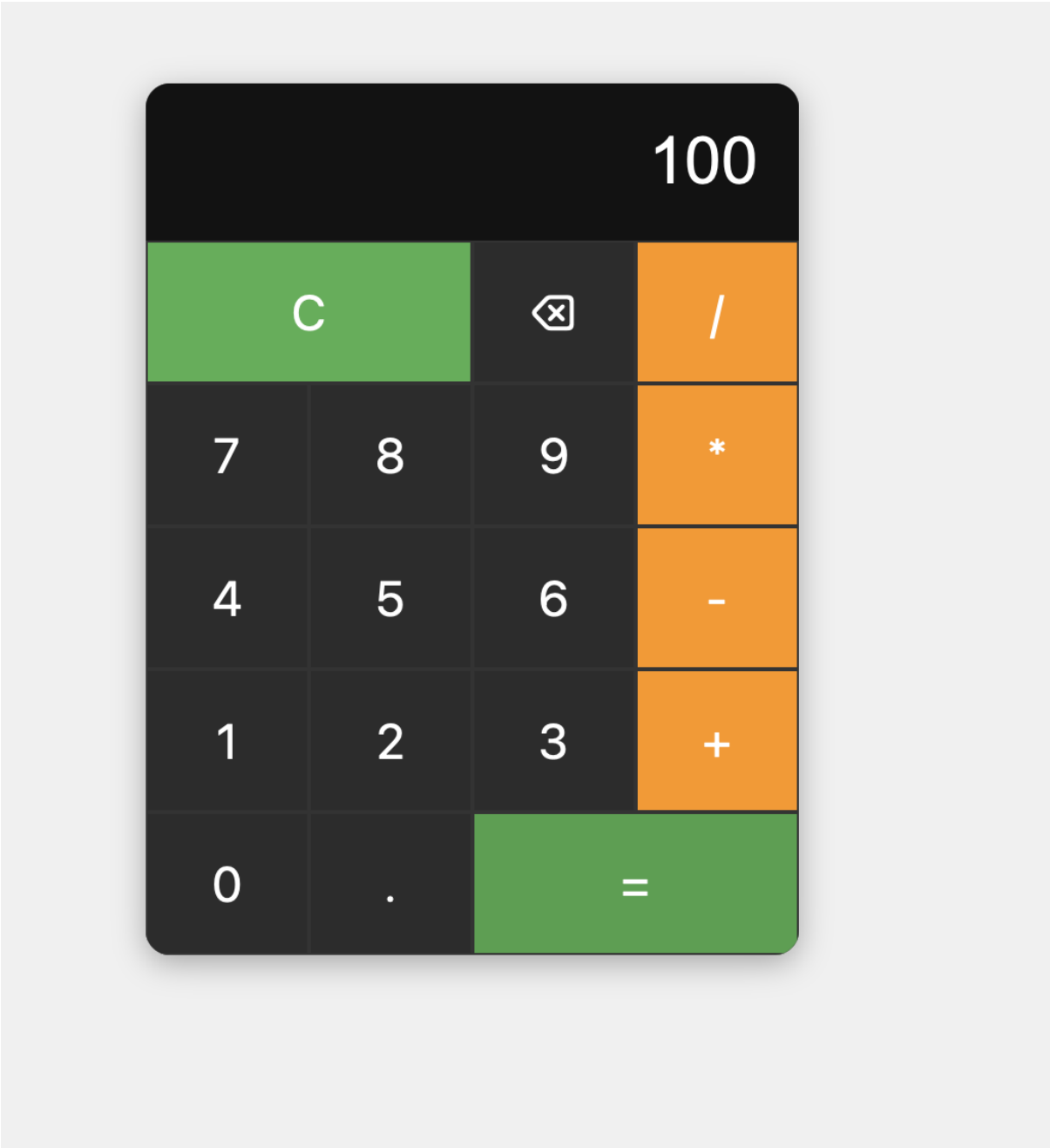

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9. GitHub Repository

https://github.com/SidharthJain2006/24BKT0106